TECHNICAL DOSSIER

Automated Foosball Table

Interactive foosball table with electronic, control, pneumatic, and future computer vision systems.

DOSSIER DATA

PROJECT LEAD
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REPOSITORY
github.com/RoboTech-URJC/Futbolin

STATUS
Technical planning

LAST UPDATE
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Project Description

The Automated Foosball Table project consists of developing an interactive foosball table that will incorporate electronic, control, pneumatic, and later computer vision systems. The proposal starts from a traditional foosball table and turns it into a technological platform for both learning and demonstrations.

The planned features include reactive LED lighting, automatic goal detection, a digital scoreboard, automatic ball relaunch, and goalkeeper automation.

Current Project Status

The project is currently in the technical planning stage. During this academic year it has not been possible to begin physical development because the funding needed to purchase the foosball table and components is still pending.

For this reason, the work carried out so far has focused on defining the project scope, organizing its development phases, and proposing the general system architecture.

Work Completed During the Year

During this year, work has focused on project conceptualization and the organization of its main modules. Specifically, the following have been defined:

- The general system structure.
- The implementation order for the features.
- The proposal for a distributed architecture with ESP32 and Raspberry Pi.
- The planned integration between sensors, lighting, scoreboard, pneumatics, and control.
- An initial approach to updating the scoreboard through MQTT.

Planned Development

Phase 1: lighting, goal detection, and scoreboard. The first phase will focus on adding LED strips, goal sensors, and a digital scoreboard.

The LEDs will be used to generate visual effects during the match, especially when a goal is scored. Goal detection will be performed with sensors placed in the goals, and that information will update the scoreboard automatically and trigger lighting animations.

Phase 2: automatic ball relaunch. Once the basic part of the game has been implemented, the ball collection and relaunch system will be developed.

After entering the goal, the ball will be collected and moved pneumatically to a raised position from which it can be launched back onto the field. This feature will add continuity to the game and reinforce the interactive nature of the project.

Phase 3: computer vision. Computer vision will be added in a later phase through an overhead camera responsible for detecting the ball position in real time.

This information will not be necessary in the first system version, but it will form the basis for more advanced autonomous functions.

Phase 4: autonomous goalkeeper. The most ambitious feature of the project will be goalkeeper automation.

Based on the detected ball position, the system will move the goalkeeper bar to place it in the most suitable area. In an initial approach, the goal will not be to create a perfect player, but rather a reactive behavior able to respond to simple game situations.

System Architecture

The project is based on a distributed architecture. Several ESP32 boards will handle specific tasks such as sensor reading, event communication, and, in advanced phases, actuator control. A Raspberry Pi will act as the central node to coordinate the system, manage the scoreboard, and process the more complex functions.

This approach enables modular development and makes it easier to progressively integrate new capabilities.

Value for the Association

The Foosball Table project has strong value as a demonstration tool for the association. It is a system that is easy to understand, visually attractive, and able to attract attention at exhibitions, events, or outreach activities.

In addition to offering an engaging experience to players, it can serve as an entry point for more people to learn about the association and become interested in other technical projects developed within RoboTech.

Next Goals

- Acquire the foosball table and planned materials.
- Begin implementing the first phase.
- Later integrate the pneumatic relaunch system.
- Develop the computer vision component.
- Begin the first tests of the autonomous goalkeeper.